



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 2 • STANDARD 6.DS.3

Describe Data by Range and Interquartile Range

Lesson 2-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

Language Objective: I can explain what the spread of a data set is using the words range, quartile, interquartile range, and vary.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Range and Interquartile Range

Range

Quartile

Interquartile range

Variability

Measure of variation



Tap any word to see what it means and a picture.

1

The range uses the values; the interquartile range uses the .

2

Subtract the least value from the value to find the range.

3

Numbers that split ordered data into four equal parts are

.

4

The spread of the middle half of the data, $Q_3 - Q_1$, is the

.

5

How spread out the numbers in a data set are is its

.

6

A single number describing how spread out a data set is is a

.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Which class scored better overall, and how do the range and the IQR show it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Range and Interquartile Range** — Two numbers that describe spread: the range covers all the data, and the interquartile range covers only the middle half.

Rango y rango intercuartílico — Dos números que describen la dispersión: el rango cubre todos los datos y el rango intercuartílico solo la mitad central.

- ☐ **Range** — The distance from the least value to the greatest value, found by subtracting: greatest – least.

Rango — La distancia del valor menor al mayor, que se halla restando: mayor – menor.

- ☐ **Quartile** — One of the three values that cut an ordered data set into four equal-sized groups.

Cuartil — Uno de los tres valores que dividen un conjunto ordenado de datos en cuatro grupos del mismo tamaño.

- ☐ **Interquartile range** — The distance across the middle half of the data, found by subtracting the first quartile from the third: $Q_3 - Q_1$.

Rango intercuartílico — La distancia a lo largo de la mitad central de los datos: se resta el primer cuartil del tercero: $Q_3 - Q_1$.

- ☐ **Variability** — How much the values in a data set differ from one another — whether they sit close together or far apart.

Variabilidad — Cuánto se diferencian entre sí los valores de un conjunto de datos: si están juntos o muy separados.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Class A's range is ____ and Class B's range is ____, so Class A's scores are more _____. Class A's IQR is ____ and Class B's IQR is ____, so the middle half of Class B is packed into a ____ stretch of points. Because 75% of Class B scored ____ or more, Class B did better ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Equal medians do not make two data sets the same — you still need to know how spread out the scores are.

Evidence: Students name a measure of spread rather than another measure of center.

Reasoning: This evidence connects the definition of range and interquartile range to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- Class A's range is ____ and Class B's range is ____, so Class A's scores are more ____.
Class A's IQR is ____ and Class B's IQR is ____, so the middle half of Class B is packed into a ____ stretch of points. Because 75% of Class B scored ____ or more, Class B did better ____.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- I know because ____.
Lo sé porque ____.
- So ____.
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Range and Interquartile Range
2. **Notes 2:** Range
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile range
5. **Notes 5:** Variability
6. **Notes 6:** Measure of variation
7. **Watch for this mistake:** *A common mistake with Range and Interquartile Range is adding the two quartiles instead of subtracting them — writing $IQR = Q3 + Q1$. With $Q1 = 42$ and $Q3 = 58$ that gives 100, which is larger than the data set's entire range of 20. Every measure of variation is a DIFFERENCE, and the IQR can never exceed the range, because the middle half is part of the whole span. If your IQR comes out bigger than your range, you added when you should have subtracted.*
8. **Try It 1:** — *Range and IQR are both measures of variation; a quartile is a marker, and variability is the idea both measures describe.*
9. **Try It 2:** A. B, because $25 - 5 = 20$ — *A's range is $18 - 12 = 6$. B's range is $25 - 5 = 20$, so B is more spread out.*